

Randomized and low-rank approximation

Query complexity from Schatten to spectral norms

Difficulty: extreme **Importance:** interesting to the community**Status:** Partially resolved

Rating rationale: Extreme because the information-theoretic transition must be uniform in norm parameter, accuracy, and dimension; community impact is the cost of low-rank approximation across error measures.

Later literature — checked 6 October 2026

The [primary arXiv abstract of Zhang et al.](#), first submitted 24 September 2026, reports

$$Q_p^* = \tilde{\Theta}(\min\{N, k \min\{p^{1/6}\varepsilon^{-1/3}, \varepsilon^{-1/2}\}\}), \quad 2 \leq p < \infty,$$

for sufficiently small ε , with universal constants, $N = \min\{m, n\}$ and success probability $2/3$. It identifies the transition $p\varepsilon \asymp 1$, with spectral endpoint $\tilde{\Theta}(\min\{N, k\varepsilon^{-1/2}\})$. For each fixed $1 \leq p < 2$, it reports $\tilde{\Theta}_p(\min\{N, k\varepsilon^{-1/3}\})$, with constants and an accuracy threshold that may depend on p .

Setting $m = n = N$ and $k = 1$ relates these results to the norm transition in this entry. **They do not determine the displayed target up to universal constant factors:** the tildes leave logarithmic factors unspecified, the fixed- p result below two is not uniform in p , and the abstract does not establish the catalog's simultaneous all-parameter guarantee with success probability $99/100$. The growing- p transition is substantial relevant progress; **status remains Partially resolved** and the original statement is unchanged. The growing-rank spectral scope is also recorded in [RA-14](#).

Evidence limit: this check retrieved the primary abstract through the search index; the full manuscript could not be retrieved. The exact theorem hypotheses and full proof have not been independently checked here, and no Lean verification is asserted for this addition.

Problem statement

For $1 \leq p < \infty$, define the Schatten norm of a real matrix M by

$$\|M\|_{\mathcal{S}_p} = \left(\sum_i \sigma_i(M)^p \right)^{1/p},$$

where $\sigma_i(M)$ are its singular values; set $\|M\|_{\mathcal{S}_\infty} = \max_i \sigma_i(M)$.

Let $n \geq 2$, $p \in [1, \infty]$, and $0 < \varepsilon < 1/2$. A randomized algorithm accesses an arbitrary unknown $A \in \mathbb{R}^{n \times n}$ only through exact products Ax or $A^T x$. Each product costs one query. Query vectors may depend on previous answers and internal randomness; computation between queries is unrestricted.

Define $q_p(n, \varepsilon)$ to be the least integer q such that some algorithm using at most q queries outputs a real unit vector v with

$$\Pr \left[\|A(I - vv^T)\|_{\mathcal{S}_p} \leq (1 + \varepsilon) \min_{\|u\|_2=1} \|A(I - uu^T)\|_{\mathcal{S}_p} \right] \geq \frac{99}{100}$$

for every A .

Open problem. Determine $q_p(n, \varepsilon)$ up to universal constant factors simultaneously in p , n , and ε , including the transition when finite p grows with n or ε^{-1} and the endpoint $p = \infty$. Finite-dimensional saturation must be included, since n column queries reveal A exactly.

This square-matrix formulation editorially fixes a two-sided oracle: the source introduction describes Av , but Algorithm 7.4 uses A and A^T . These models coincide on its symmetric lower-bound inputs. The relative error is in the norm, not its p th power.

Why it matters in numerical linear algebra

This asks how the matrix-product cost changes between aggregate singular-value error and worst-direction error.

References

1. Ainesh Bakshi and Shyam Narayanan, *Krylov Methods are (nearly) Optimal for Low-Rank Approximation*, arXiv:2304.03191v1 (2023), Open Question 1.11; Theorems 1.1 and 1.5, Algorithm 7.4.
2. Praneeth Kacham and David P. Woodruff, *Faster Algorithms for Schatten- p Low Rank Approximation*, APPROX/RANDOM 2024, article 55: later running-time bounds.
3. H. Zhang, W. Wu, C. Zhang, Y. Li, C. Zheng, C. Fang, H. Li, and Z. Lin, *Matrix-Vector Complexity of Low-Rank Approximation*, arXiv:2609.35840, 24 September 2026, abstract; the present literature notice is limited to that primary abstract.

Status check

For fixed finite p , the source gives a dimension-independent query bound; the spectral lower bound grows with $\log n$ in its stated dimension regime. Neither gives optimal uniform dependence on growing p . The 2024 work improves running time.

Checked 2026-09-08: both arXiv records list only v1. Searches for “Open Question 1.11,” Schatten phase transition, query complexity, and 2025/2026 found no resolution. The older rank-one spectral lower-bound question was solved by the 2023 source. This is a bounded check.

Audit — 2026-09-10

Rechecked Bakshi–Narayanan, Theorem 1.1 and Open Question 1.11. The spectral endpoint has matching bounds in its stated dimension regime; optimal growing- p dependence remains open. Later Schatten-query searches found no full characterization; finite- p near-optimal results and running-time improvements leave this transition unresolved.